

A.5 Relativity [HL only]
Topic warm-up · IB Physics (2023 syllabus)
with worked answers

Name: _____

Date: _____

Total: **25 marks**

$c = 3.0 \times 10^8 \text{ m s}^{-1}$. Show your working.

Question 1 — The interstellar courier (11 marks)

A courier ship travels at $0.80c$ to a station 12 light years away (as measured from Earth).

- (a) State the two postulates of special relativity. [2]
- (b) Show that the Lorentz factor for the journey is $\gamma = \frac{5}{3}$. [2]
- (c) The pilot's clock records 9.0 years for the trip. Calculate the journey time recorded on Earth, and state whose measurement is the *proper time*. [3]
- (d) Calculate the Earth–station distance as measured by the pilot, and explain why she can consistently believe she arrived “early”. [4]

Question 2 — The overtaking starships (5 marks)

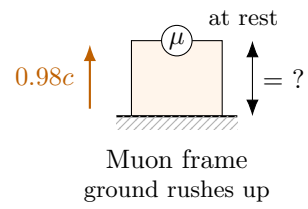
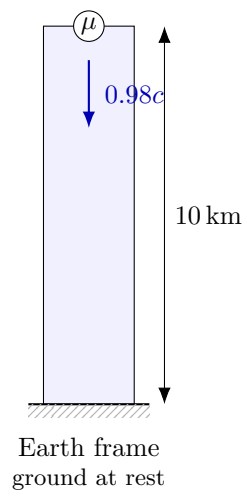
Two starships race along the same straight line, measured from a space station: ship P at $0.60c$ and ship Q at $0.80c$.

- (a) A cadet computes Q's speed relative to P as $0.80c - 0.60c = 0.20c$. Name the (classical) rule the cadet has used, and state when it is an acceptable approximation. [2]
- (b) Using $u' = \frac{u - v}{1 - uv/c^2}$, calculate the actual relative speed. [3]

Question 3 — The muons that should not arrive (9 marks)

Muons created about 10 km up in the atmosphere travel towards the ground at $0.98c$ ($\gamma \approx 5.0$). Their proper (rest-frame) half-life corresponds to a mean lifetime of about $2.2 \mu\text{s}$.

- (a) Show that, without relativity, a typical muon would travel only about 650 m before decaying. [2]
- (b) Explain the muons' arrival at the ground from the *Earth frame*, with a supporting calculation. [3]
- (c) Explain the same arrival from the *muon's frame*, with a supporting calculation. [3]
- (d) State what this experiment demonstrates about time dilation. [1]



Answers

Question 1 — The interstellar courier

- (a) (1) The laws of physics are identical in all inertial frames. (2) The speed of light in vacuum is the same for all inertial observers, regardless of the motion of the source or the observer.
- (b) $\gamma = \frac{1}{\sqrt{1-0.80^2}} = \frac{1}{\sqrt{0.36}} = \frac{1}{0.60} = \frac{5}{3}$.
- (c) Earth time $= \gamma \times 9.0 = \frac{5}{3} \times 9.0 = 15$ years. (Check: 12 ly at $0.80c$ takes $12/0.80 = 15$ y.) The *pilot* records the proper time, because both events, departure and arrival, happen at her own location.
- (d) $L' = 12/\gamma = 12 \times \frac{3}{5} = 7.2$ ly. In her frame the distance itself is contracted, so covering 7.2 ly at $0.80c$ takes $7.2/0.80 = 9.0$ y, exactly what her clock reads. Both frames agree on every clock reading; each simply has its own consistent account of why.

Question 2 — The overtaking starships

- (a) Galilean velocity addition. It is an acceptable approximation when both speeds are much less than c .
- (b) $u' = \frac{0.80c - 0.60c}{1 - (0.80)(0.60)} = \frac{0.20c}{0.52} = 0.38c$.

Question 3 — The muons that should not arrive

- (a) $d = 0.98 \times 3.0 \times 10^8 \times 2.2 \times 10^{-6} = 650$ m, far short of the 10 km they must cross.
- (b) *Earth frame.* The muon's lifetime is dilated to $\gamma \times 2.2 = 11 \mu\text{s}$. The journey takes $10\,000/(0.98 \times 3.0 \times 10^8) = 34 \mu\text{s}$, which is $34/11 \approx 3.1$ dilated lifetimes. A real fraction therefore survives, $e^{-3.1} \approx 5\%$, rather than the essentially none implied by (a).
- (c) *Muon frame.* Its own clock runs normally, giving the proper $2.2 \mu\text{s}$, but the atmosphere is length-contracted to $10/5.0 = 2.0$ km. That depth sweeps past in $2000/(0.98 \times 3.0 \times 10^8) = 6.8 \mu\text{s}$, again ≈ 3.1 proper lifetimes, so the same $\approx 5\%$ survives. The two frames disagree about *which* quantity changes, and agree exactly on the observable muon count.
- (d) That time dilation and length contraction are real, measurable effects rather than illusions: the muon count at sea level matches the relativistic prediction and contradicts the classical one.